Harsha Koushik Teja Aila

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Education

Arizona State University — Aug 2025 – May 2027
M.S. in Data Science, Analytics and Engineering — *Tempe, AZ*
- **Relevant Coursework:** Data Processing at Scale, ML, Statistical ML, Data Mining, Statistical Data Analysis

CVR College of Engineering — Dec 2021 – Apr 2025
B.Tech in Computer Science (CGPA: 8.8/10.0) — *Hyderabad, India*

Technical Skills

Languages: Python, Java, C, C#, SQL, Apex, JavaScript, HTML/CSS
Frameworks: React, Node.js, REST APIs, Tailwind CSS, Lightning Web Components
ML / Data: PyTorch, PyTorch Geometric, scikit-learn, Pandas, NumPy, NetworkX, gensim
Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Kubernetes, Kafka, CI/CD, GitHub Actions, Linux
Databases: PostgreSQL, MongoDB, Neo4j
Tools: Git, VS Code, IntelliJ, PyCharm

Experience

Salesforce Developer Intern — Nov 2023 – Jan 2024
SmartInternz (Salesforce Partner) — *Remote*
- Engineered 15+ server-side workflows and batch jobs in Apex (Java-based syntax), improving system processing efficiency by 35% across business automation pipelines.
- Developed 8+ modular front-end components in JavaScript (Lightning Web Components) and integrated 5+ REST APIs, reducing manual data handling by 40%.
- Configured CI/CD pipelines using GitHub Actions for automated builds, testing, and deployment, enabling 20% faster release cycles across environments.
- Achieved 90%+ unit test coverage through test-driven development, reducing post-deployment defects by 30%.

Projects

Multi-Label Node Classification: Classical Embeddings vs GNNs | *Python, PyTorch Geometric, scikit-learn, gensim*
- Led a 6-person team to implement DeepWalk and Node2Vec from scratch for multi-label node classification across 3 benchmark graphs (BlogCatalog, PPI, Wikipedia), achieving a 99% Micro-F1 improvement over the degree baseline on BlogCatalog via a combined 256-dim embedding.
- Independently extended the project into a 7-method benchmark by adding GCN, GraphSAGE, and GAT in PyTorch Geometric with 5-seed statistical validation, surfacing GAT training instability (± 0.151 variance) hidden by single runs.
- Profiled compute cost across all GNNs, quantifying that standard GAT required 84× more training time and 27× more GPU memory than GraphSAGE on dense featureless graphs, informing model-selection guidance.

Distributed Data Processing Pipeline | *Kafka, Neo4j, Docker, Kubernetes*
- Engineered a distributed streaming pipeline ingesting real-time trip data using Kafka producers/consumers and a Neo4j graph database, supporting 100K+ records with sub-second processing latency.
- Containerized the full stack with Docker and orchestrated multi-container deployment on Kubernetes (Minikube), enabling fault-tolerant, reproducible execution across environments.
- Optimized Cypher graph queries and Kafka partitioning strategy to improve throughput, earning a 60/60 graded evaluation for production-grade distributed-systems design.

M.Assist – Voice Assistant | *Python, Vosk, Whisper, sentence-transformers*
- Architected a voice assistant with a three-tier pipeline: wake-word detection, two-tier speech-to-text (fast path + fallback), and a hybrid rules-plus-embedding intent router.
- Routed 70%+ of intents to local handlers and integrated external LLM APIs for complex queries, eliminating cloud dependency for sensitive operations and reducing average response latency by 40% vs. a cloud baseline.
- Built a modular, YAML-configured handler system in Python supporting 15+ command categories with structured logging and a CLI test harness.

EState – Full-Stack Real Estate Platform | *Node.js, Express, MongoDB, JWT*
- Built a secure full-stack platform with JWT authentication and role-based access control across 3 user tiers, exposing 15+ REST APIs on Node.js/Express.
- Designed and indexed a MongoDB schema for users, listings, and transactions, optimizing backend workflows to reduce query response latency by 60%.
- Integrated automated contact handling and notification workflows, eliminating manual intervention across 100% of inbound leads.